Nonparametric estimation of causal parameters

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1 Unconfoundedness

Without loss of generality, let $f(1,W)$ and $f(0,W)$ be the conditional expectation of the potential outcomes $Y(1), Y(0)$ given W. We may write

$$\begin{aligned} Y(1) &= f(1,W) + \varepsilon(1) \\ Y(0) &= f(0,W) + \varepsilon(0) \end{aligned}$$

where

$$\begin{aligned} \mathbb{E}[\varepsilon(1)|W] &= 0 \\ \mathbb{E}[\varepsilon(0)|W] &= 0 \end{aligned}$$

Suppose that the unconfoundedness assumption hold so that the assignment to treatment $V \in \{1,0\}$ can be viewed as random, conditional on the observations W.

$$\{Y(1), Y(0)\} \perp V|W \tag{1}$$

Given W, the randomness in $Y(1)$ and $Y(0)$ comes only from $\varepsilon(1)$ and $\varepsilon(0)$ so that the unconfoundedness assumption implies

$$\{\varepsilon(1), \varepsilon(0)\} \perp V|W$$

This implies that both the conditional expectation errors $\varepsilon(1), \varepsilon(0)$ have mean zero even after conditioning on both V and W.

$$\begin{aligned} \mathbb{E}[\varepsilon(1)|V,W] &= \mathbb{E}[\varepsilon(1)|W] = 0 \\ \mathbb{E}[\varepsilon(0)|V,W] &= \mathbb{E}[\varepsilon(0)|W] = 0 \end{aligned}$$

The fundamental problem of causal inference refers to the fact that we cannot observe the two potential outcomes of each individuals. Let Y be the outcome after assignment. Then, we may write

$$\begin{aligned} Y &= VY(1) + (1-V)Y(0) \\ &= f(V,W) + \varepsilon \end{aligned}$$

where

$$\varepsilon = V\varepsilon(1) + (1-V)\varepsilon(0)$$

The unconfoundedness assumption implies that

$$\begin{aligned}\mathbb{E}[\varepsilon|V,W] &= \mathbb{E}[V\varepsilon(1) + (1-V)\varepsilon(0)|V,W] \\ &= V\mathbb{E}[\varepsilon(1)|V,W] + (1-V)\mathbb{E}[\varepsilon(0)|V,W] \\ &= 0\end{aligned}$$

This implies that we may regard $f(V,W)$ as the conditional expectation of the observable Y given V,W. Thus, we may perform any nonparametric regression of Y on V,W to get an estimate of the conditional expectation $\hat{f}(V,W)$. Let the conditional average treatment effect (CATE) be defined as

$$\begin{aligned}\tau(W) &= \mathbb{E}[Y(1) - Y(0)|W] \\ &= f(1,W) - f(0,W)\end{aligned}$$

Provided that we use a consistent estimator for $f(V,W)$, we may obtain an estimate for CATE as

$$\hat{\tau}(W) = \hat{f}(1,W) - \hat{f}(0,W)$$

Let the average treatment effect (ATE) be defined as

$$\begin{aligned}\tau &= \mathbb{E}[Y(1) - Y(0)] \\ &= \mathbb{E}[f(1,W) - f(0,W)]\end{aligned}$$

We may estimate the ATE as

$$\hat{\tau} = \frac{1}{n}\sum_{i=1}^{n}(\hat{f}(1,W_i) - \hat{f}(0,W_i))$$

Let the average treatment effect on treated (ATT) be defined as

$$\begin{aligned}\tau_T &= \mathbb{E}[Y(1) - Y(0)|V=1] \\ &= \mathbb{E}[f(1,W) - f(0,W)|V=1]\end{aligned}$$

Then, we may estimate the ATT as

$$\hat{\tau}_T = \frac{1}{n_1}\sum_{i\in I_1}(\hat{f}(1,W_i) - \hat{f}(0,W_i))$$

where I_1 denotes the index set of treated individuals and n_1 denotes the number of treated individuals.

1.1 CATE on $W^{(1)}$

Suppose we split W into $(W^{(1)}, W^{(2)})$ and focus on the average treatment effect conditional on only $W^{(1)}$.

$$\begin{aligned}\tau(w^{(1)}) &= \mathbb{E}[Y(1) - Y(0)|W^{(1)} = w^{(1)}] \\ &= \mathbb{E}[f(1,W) - f(0,W)|W^{(1)} = w^{(1)}]\end{aligned}$$

Here,

$$
\begin{aligned}
\mathbb{E}[f(1,W)|W^{(1)}=w^{(1)}] &= \mathbb{E}[f(1,W^{(1)},W^{(2)})|W^{(1)}=w^{(1)}] \\
&= \int f(1,w^{(1)},w^{(2)})p(W^{(2)}=w^{(2)}|W^{(1)}=w^{(1)})dw^{(2)}
\end{aligned}
$$

Suppose we use a kernel estimator for the conditional density $p(w^{(2)}|w^{(1)})$.

$$
\begin{aligned}
\hat{p}(w^{(1)},w^{(2)}) &= \frac{1}{nh_1h_2}\sum_{j=1}^{n} K\left(\frac{W_j^{(1)}-w^{(1)}}{h_1}\right) K\left(\frac{W_j^{(2)}-w^{(2)}}{h_2}\right) \\
\hat{p}(w^{(1)}) &= \frac{1}{nh_1}\sum_{j=1}^{n} K\left(\frac{W_j^{(1)}-w^{(1)}}{h_1}\right)
\end{aligned}
$$

Then, the estimated conditional expectation is given by

$$
\hat{\mathbb{E}}[f(1,W)|W^{(1)}=w^{(1)}] = \int f(1,w^{(1)},w^{(2)})\hat{p}(W^{(2)}=w^{(2)}|W^{(1)}=w^{(1)})dw^{(2)}
$$

$$
= \frac{1}{\sum_{j=1}^{n} K\left(\frac{W_j^{(1)}-w^{(1)}}{h_1}\right)} \int f(1,w^{(1)},w^{(2)})\frac{1}{h_2}\sum_{j=1}^{n} K\left(\frac{W_j^{(1)}-w^{(1)}}{h_1}\right) K\left(\frac{W_j^{(2)}-w^{(2)}}{h_2}\right) dw^{(2)}
$$

$$
= \frac{1}{\sum_{j=1}^{n} K\left(\frac{W_j^{(1)}-w^{(1)}}{h_1}\right)} \frac{1}{h_2}\sum_{j=1}^{n} K\left(\frac{W_j^{(1)}-w^{(1)}}{h_1}\right) \int f(1,w^{(1)},w^{(2)}) K\left(\frac{W_j^{(2)}-w^{(2)}}{h_2}\right) dw^{(2)}
$$

Using change of variables $\frac{W_j^{(2)}-w^{(2)}}{h_2} = u$,

$$
\hat{\mathbb{E}}[f(1,W)|W^{(1)}=w^{(1)}] = \frac{1}{\sum_{j=1}^{n} K\left(\frac{W_j^{(1)}-w^{(1)}}{h_1}\right)} \sum_{j=1}^{n} K\left(\frac{W_j^{(1)}-w^{(1)}}{h_1}\right) \int f(1,w^{(1)},W_j^{(2)}-h_2u)K\left(u\right)du
$$

Using second order Taylor expansion of $f(1,w^{(1)},W_j^{(2)}-h_2u)$ at $u=0$, we may write

$$
f(1,w^{(1)},W_j^{(2)}-h_2u) \approx f(1,w^{(1)},W_j^{(2)}) - h_2f'(1,w^{(1)},W_j^{(2)})u + h_2^2f''(1,w^{(1)},W_j^{(2)})u^2
$$

where f' and f'' denotes the first and second derivative w.r.t. the third argument. Assuming we are using a second order kernel, and that $h_2^2 \approx 0$, we have

$$
\hat{\mathbb{E}}[f(1,W)|W^{(1)}=w^{(1)}] \approx \frac{\sum_{j=1}^{n} K\left(\frac{W_j^{(1)}-w^{(1)}}{h_1}\right) f(1,w^{(1)},W_j^{(2)})}{\sum_{j=1}^{n} K\left(\frac{W_j^{(1)}-w^{(1)}}{h_1}\right)}
$$

Similarly, we have

$$
\hat{\mathbb{E}}[f(0,W)|W^{(1)}=w^{(1)}] \approx \frac{\sum_{j=1}^{n} K\left(\frac{W_j^{(1)}-w^{(1)}}{h_1}\right) f(0,w^{(1)},W_j^{(2)})}{\sum_{j=1}^{n} K\left(\frac{W_j^{(1)}-w^{(1)}}{h_1}\right)}
$$

Here, both $W^{(1)}$ and $W^{(2)}$ were treated as univariate random variables. Extension to multidimensional case is straightforward using multivarite kernel function.